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- **LED:** Lower power draw and longer life (~20,000 h) but often lower peak brightness, limiting large-screen performance.
- **Laser:** Combines high brightness, instant on/off, and 20,000+ h lifespan; ideal for ultra-short-throw and premium home cinema models.



2) Display panels: DLP vs. LCD vs. LCoS

- **DLP (Digital Light Processing):** Sharp images and compact form factors; may exhibit “rainbow” artifacts under certain conditions.
- **LCD (3LCD):** Rich color and no rainbow effect; contrast ratios lag behind DLP and LCoS unless paired with advanced dimming.
- **LCoS (Liquid Crystal on Silicon):** Reflective like DLP but with LCD’s color fidelity, achieving the best on/off contrast and smooth images – often in high-end models.

Practical recommendation: For budget setups in controlled-light rooms, DLP offers value. For mixed lighting or color-critical viewing, LCoS or premium LCD projectors excel. Laser light sources enhance longevity and color stability across all panel types.

Throw distance and screen size

Accurate throw-distance calculations ensure your projector lens fills the screen fully without distortion.

1) Online projection calculators

Web tools like ProjectorCentral’s Projection Calculator Pro let you enter lens specifications and desired screen width to compute required projector-to-screen distance. Epson and JVC also offer proprietary lens calculators for their models.

2) Manual throw ratio math

- $\text{Throw Ratio} = (\text{Distance from lens to screen}) \div (\text{Image width})$
- For a throw ratio of 1.2:1, a 2 m-wide screen requires 2.4 m of distance.

Step-by-step example:

1. **Measure Available Space:** Your home theater wall is 3 m from the mounting point.

2. **Decide Screen Width:** At a 16:9 aspect ratio for 100" diagonal, width ≈ 2.21 m.
3. **Compute Effective Throw Ratio:** $3 \text{ m} \div 2.21 \text{ m} \approx 1.36:1$.
4. **Select Projector:** Choose a model with adjustable lenses covering 1.20–1.50 throw ratios, ensuring flexibility.

Brightness requirements

1) ANSI vs. LED lumens

Manufacturers rate lamps in ANSI lumens, while LED/laser models often quote “LED lumens,” which measure perceived brightness differently – typically ~20–30% lower than ANSI spec for equivalence.

2) Matching lumens to ambient light

- **Dark Room (Controlled Lighting):** 1,000–2,500 ANSI lumens suffice for 100–150" screens.
- **Occasional Lights On:** Aim for 2,300–4,000 lumens; higher if screens exceed 120".
- **Living Rooms with Windows:** Consider 4,000+ lumens and an ALR screen to combat washout.

Tip: Measure foot-lamberts (ftL) on your chosen screen surface – 16–26 ftL offers good dark-room performance; 27–39 ftL is ideal for low ambient light; 40+ ftL for brighter rooms.

Projection Surfaces

1) Screen material types

- **Matte White:** Standard gain (1.0), wide viewing angles, best in dark rooms.
- **High-Gain:** Gain 1.3–2.0 for brighter images in controlled settings, but narrower viewing cones and hotspotting.
- **Grey/High Contrast:** Improves black levels in moderate ambient light by absorbing stray light.

2) Ambient light rejecting (ALR) screens

ALR screens use micro-louvered or reflective coatings to bounce projector light back to the viewer while rejecting room light.

